Neural Networks - Task 2

Bird classification using backpropagation

1. Introduction

The aim of this task was to classify three penguin species using a multilayer neural network trained with backpropagation. Sigmoid and tanh were compared using the same dataset. The network uses five input features and three output neurons, with one output for each species.

2. Dataset and experiment setup

The dataset contains 50 samples for each of Adelie, Chinstrap and Gentoo. The first 30 samples of each species were used for training and the remaining 20 for testing, giving 90 training samples and 60 test samples. Missing numerical values were filled using training-set means, and all features were standardized using training-set statistics. OriginLocation was encoded as Biscoe = 0, Dream = 1 and Torgersen = 2.

A total of 48 configurations were tried, using 1, 2, 3, 4, 5 hidden layers. The learning rates were 0.01, 0.05, and the epoch counts were 600, 1000. Weights and enabled biases were initialized with small random values between -0.1 and 0.1.

To compare the settings, 24 of the 30 training samples per species were used to fit each network and the other 6 were used for validation. Preprocessing was fitted on those 72 training samples. The configuration with the highest validation accuracy was selected for each activation; lower validation loss was used to break ties. Each selected network was then retrained on all 90 training samples before testing on the remaining 60.

3. Results

Activation	Train accuracy	Test accuracy	LR	Epochs	Hidden layers	Neurons per layer	Bias	Seed
Sigmoid	98.89%	98.33%	0.01	600	1	12	On	42
Tanh	100.00%	96.67%	0.05	1000	2	12, 8	On	42

The sigmoid network classified 59 out of 60 test samples correctly (98.33%). The tanh network classified 58 out of 60 test samples correctly (96.67%).

4. Effect of the number of hidden layers

The best validation result for each number of hidden layers is listed below. These accuracies were calculated on the 18 validation samples. The final test results for the selected networks are given in the previous table.

Activation	Hidden layers	Neurons per layer	LR	Epochs	Validation accuracy	Validation loss
Sigmoid	1	12	0.01	600	88.89%	0.06973
Sigmoid	2	12, 8	0.01	1000	88.89%	0.14173
Sigmoid	3	8, 8, 5	0.01	1000	33.33%	0.33333
Sigmoid	4	8, 8, 5, 3	0.01	1000	33.33%	0.33333
Sigmoid	5	12, 10, 8, 6, 4	0.01	1000	33.33%	0.33333
Tanh	1	12	0.01	600	88.89%	0.10838
Tanh	2	12, 8	0.05	1000	88.89%	0.08668
Tanh	3	8, 8, 5	0.05	1000	88.89%	0.10197
Tanh	4	8, 8, 5, 3	0.01	1000	88.89%	0.10964
Tanh	5	12, 10, 8, 6, 4	0.05	1000	83.33%	0.13453

Discussion

The table shows that increasing the number of hidden layers did not consistently improve validation accuracy. Several configurations gave the same accuracy, so validation loss helped choose between them. These results apply to the settings tested here; changing the initialization, learning rate or number of epochs could change how the deeper networks learn. The selected numbers of hidden layers were 1 for sigmoid and 2 for tanh.

Sigmoid results

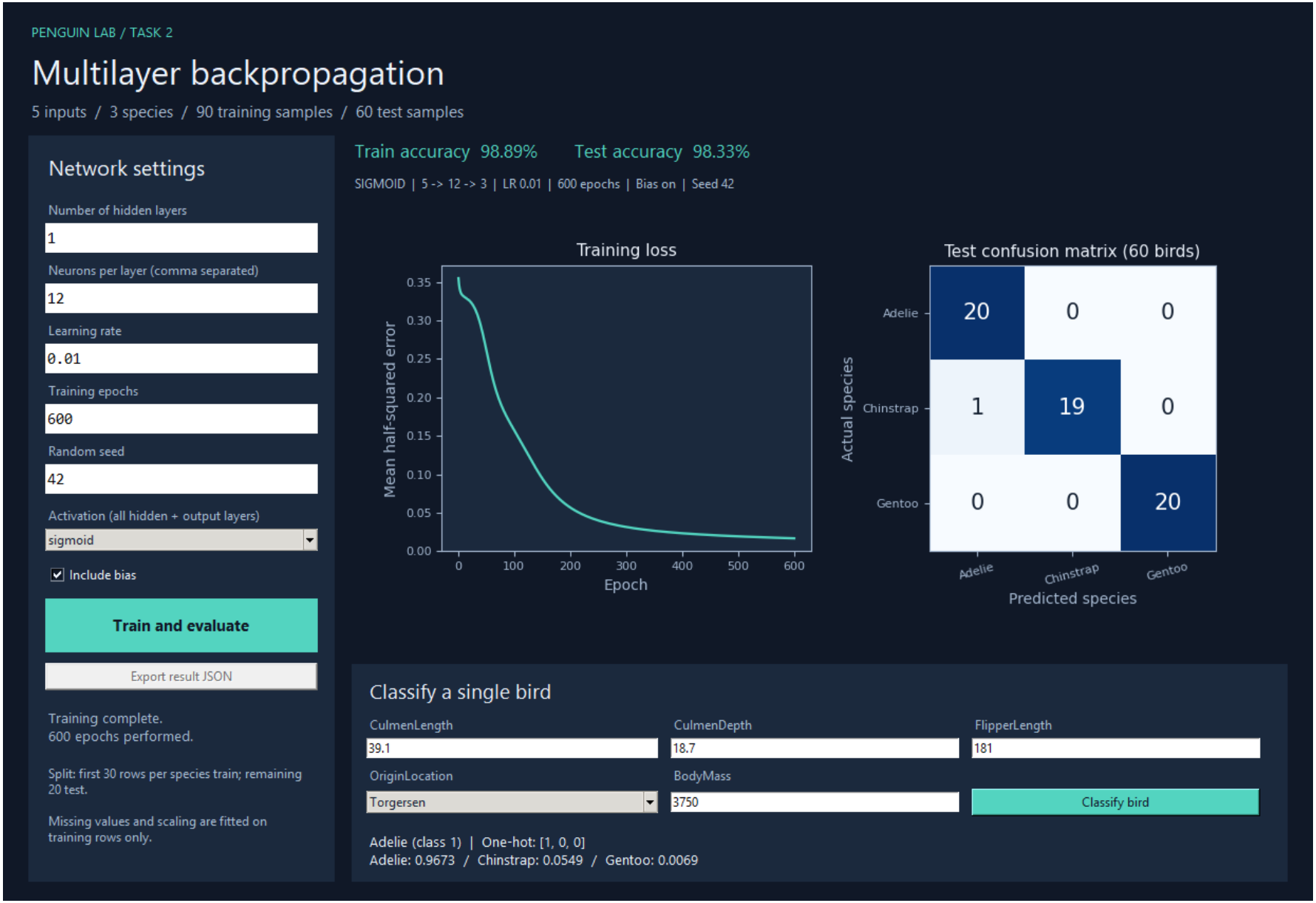


Figure 1. Settings, training loss and test confusion matrix for the sigmoid network.

Tanh results

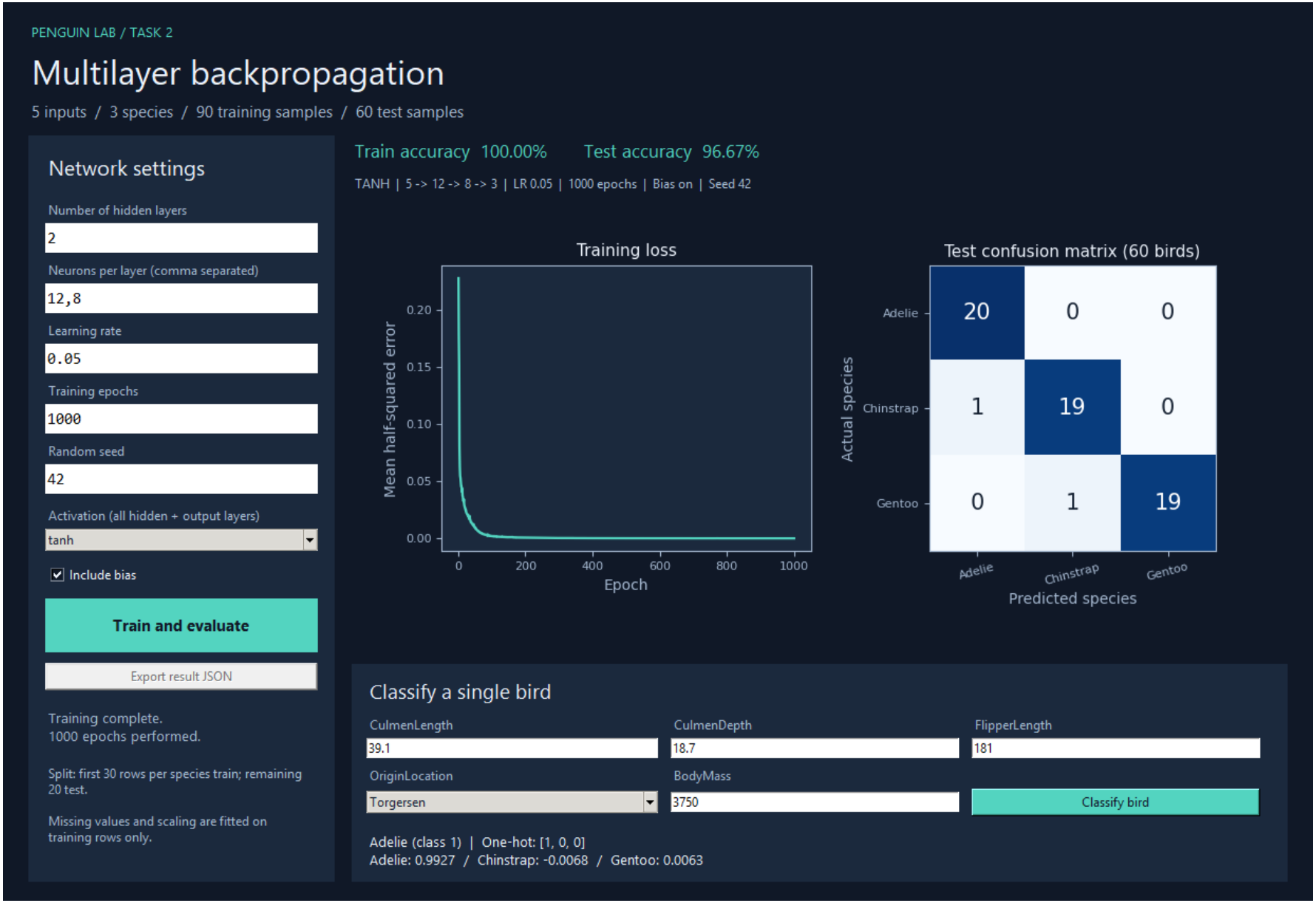


Figure 2. Settings, training loss and test confusion matrix for the tanh network.